

S29 — Probability: Bayes + Conditional + Business Revision

1 | Bayes — Customer Conversion

Traffic Source	Visitors	Purchased
Advertisement	800	120
Organic Search	1,200	180

1. Total visitors.
 2. Total purchasers.
 3. $P(\text{Purchased})$.
 4. $P(\text{Purchased} \mid \text{Advertisement})$.
 5. $P(\text{Advertisement} \mid \text{Purchased})$.
 6. Which probability is higher: Q4 or Q5?
 7. Explain why Q4 and Q5 are different.
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2 | Conditional Probability — Support Requests

Channel	Requests	Resolved
Chat	900	720
Email	600	360

1. Total requests.
2. Total resolved.
3. $P(\text{Resolved} \mid \text{Chat})$.
4. $P(\text{Resolved} \mid \text{Email})$.
5. $P(\text{Chat} \mid \text{Resolved})$.
6. Which channel has the higher resolution rate?
7. Which channel contributes more resolved requests?

8. Are Q3 and Q5 asking the same question?

3 | Mutually Exclusive vs Overlapping — Revision

A standard six-sided die is rolled.

- Event A = getting an even number.
- Event B = getting an odd number.
- Event C = getting a number greater than 3.

1. Write Event A.
 2. Write Event B.
 3. Are A and B mutually exclusive?
 4. Write $A \cap B$.
 5. Are A and C mutually exclusive?
 6. Write $A \cap C$.
 7. Explain why A and C overlap.
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4 | Addition Rule — Overlapping Events

A company has 100 customers:

- 60 use the mobile app.
 - 45 use the website.
 - 25 use both.
1. Find $P(App)$.
 2. Find $P(Website)$.
 3. Find $P(App \cap Website)$.
 4. Find $P(App \cup Website)$.
 5. How many customers use at least one of the two channels?
 6. Why can't we simply add $60 + 45$?
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5 | Complement Rule — Revision

A company finds that the probability of a customer cancelling a subscription is **8%**.

1. Find the probability that a customer does **not** cancel.
 2. Convert the answer to percentage.
 3. Does 92% mean exactly 92 out of every 100 customers will definitely stay?
 4. Explain the difference between **probability** and **guarantee**.
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6 | Bayes — Fraud Detection

Transaction Type	Transactions	Fraud
Online	9,000	180
Offline	3,000	60

1. Total transactions.
 2. Total fraud transactions.
 3. $P(\text{Fraud} \mid \text{Online})$.
 4. $P(\text{Fraud} \mid \text{Offline})$.
 5. $P(\text{Online} \mid \text{Fraud})$.
 6. Which has the higher fraud rate?
 7. If a transaction is known to be fraudulent, what does the denominator represent in Q5?
 8. Can we conclude that every online transaction is fraudulent?
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7 | Conversion Rate — Marketing Revision

Campaign	Customers Reached	Purchased
A	5,000	400
B	2,000	300

1. Conversion rate of Campaign A.
2. Conversion rate of Campaign B.

3. Which campaign has the higher conversion rate?
 4. Which campaign generated more purchasers?
 5. Can we say Campaign A performed better just because it generated more purchasers?
 6. Explain **conversion rate vs number of purchasers**.
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8 | Find the Wrong Analyst Interpretation

An analyst says:

“The conversion rate is 20%. Therefore, if 500 customers are reached, exactly 100 customers will purchase.”

1. Is the conclusion correct?
 2. What is wrong with the word “**exactly**”?
 3. What does 20% represent?
 4. What number would we expect approximately from 500 customers?
 5. Is that number guaranteed?
 6. Rewrite the analyst's conclusion correctly.
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9 | Three-Group Bayes — Employee Skills

Department	Employees	Python Skilled
Data Analytics	200	120
Finance	300	60
Marketing	500	50

1. Total employees.
2. Total Python-skilled employees.
3. $P(\text{Python Skilled})$.
4. $P(\text{Python Skilled} \mid \text{Data Analytics})$.
5. $P(\text{Data Analytics} \mid \text{Python Skilled})$.
6. Which department has the highest Python skill **rate**?

7. Which department has the largest number of Python-skilled employees?
 8. Are Q6 and Q7 asking the same thing?
 9. Explain the difference between **rate** and **count**.
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10 | Full Data Analyst Case — Churn + Bayes + Business Decision

A subscription company has:

Plan	Customers	Churned
Basic	3,000	450
Premium	1,500	90
Enterprise	500	25

1. Total customers.
2. Total churned customers.
3. Overall churn probability.
4. $P(\text{Churned} \mid \text{Basic})$.
5. $P(\text{Churned} \mid \text{Premium})$.
6. $P(\text{Churned} \mid \text{Enterprise})$.
7. $P(\text{Basic} \mid \text{Churned})$.
8. Which plan has the **highest churn rate**?
9. Which plan has the **largest number of churned customers**?
10. Are Q8 and Q9 asking the same question?
11. Which plan should be investigated if the goal is to reduce the **highest churn rate**?
12. Which plan should be investigated if the goal is to reduce the **largest number of churned customers**?
13. Can management conclude that the Basic plan is causing churn?
14. Give **two additional pieces of information** you would investigate before making a final business decision.

15. Explain why **number of customers, churn rate, and number of churned customers** should not be treated as the same metric.